

ETHANOL (ANHYDROUS)**ICSC: 0044 (May 2018)**

Ethyl alcohol
Absolute ethanol
Methyl carbinol
Grain alcohol

CAS #: 64-17-5**UN #: 1170****EC Number: 200-578-6**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Highly flammable. Vapour/air mixtures are explosive. Risk of fire and explosion on contact with incompatible substances. See Chemical Dangers.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting. Do NOT use compressed air for filling, discharging, or handling. NO contact with incompatible materials: See Chemical Dangers	Use water spray, powder, alcohol-resistant foam, carbon dioxide. In case of fire: keep drums, etc., cool by spraying with water.

STRICT HYGIENE! PREVENT GENERATION OF MISTS!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Headache. Fatigue. Drowsiness.	Use ventilation, local exhaust or breathing protection.	Fresh air, rest.
Skin	Dry skin.	Protective clothing. Apron. Protective gloves.	Remove contaminated clothes. Rinse skin with plenty of water or shower.
Eyes	Redness. Pain. Burning sensation.	Wear safety goggles.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion	Burning sensation. Headache. Confusion. Dizziness. Unconsciousness.	Do not eat, drink, or smoke during work.	Rinse mouth. Give one or two glasses of water to drink. Refer immediately for medical attention.

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Remove all ignition sources. Ventilation. Do NOT wash away into sewer. Collect leaking and spilled liquid in covered containers as far as possible. Absorb remaining liquid in inert absorbent. Wash away remainder with plenty of water. Store and dispose of according to local regulations.	According to UN GHS Criteria  DANGER Highly flammable liquid and vapour Harmful if swallowed Causes serious eye irritation May cause damage to organs through prolonged or repeated exposure Transportation UN Classification UN Hazard Class: 3; UN Pack Group: II
STORAGE	
Fireproof. Separated from : see Chemical Dangers.	
PACKAGING	

ETHANOL (ANHYDROUS)**ICSC: 0044****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS LIQUID WITH CHARACTERISTIC ODOUR.

Physical dangers

The vapour mixes well with air, explosive mixtures are easily formed.

Chemical dangers

Reacts slowly with calcium hypochlorite, silver oxide and ammonia. This generates fire and explosion hazard. Reacts violently with strong oxidants such as nitric acid, silver nitrate, mercuric nitrate and magnesium perchlorate. This generates fire and explosion hazard.

Formula: $\text{CH}_3\text{CH}_2\text{OH}$ / $\text{C}_2\text{H}_6\text{O}$

Molecular mass: 46.1

Boiling point: 78°C

Melting point: -114 °C

Relative density (water = 1): 0.79

Solubility in water: miscible

Vapour pressure, kPa at 20°C: 5.8

Relative vapour density (air = 1): 1.6

Relative density of the vapour/air-mixture at 20°C (air = 1): 1.03

Flash point: 12.0 °C c.c.

Auto-ignition temperature: 400°C

Explosive limits, vol% in air: 3.1-27.7

Octanol/water partition coefficient as log Pow: -0.32

Viscosity: 1.074 mPa*s at 20°C

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation of its vapour and by ingestion.

Effects of short-term exposure

The substance is severely irritating to the eyes. The vapour at high levels is irritating to the eyes and respiratory tract. The substance may cause effects on the central nervous system.

Inhalation risk

A harmful contamination of the air will be reached rather slowly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure

The substance defats the skin, which may cause dryness or cracking. The substance may have effects on the upper respiratory tract and central nervous system. This may result in irritation, headache, fatigue and lack of concentration. See Notes.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 1000 ppm as STEL; A3 (confirmed animal carcinogen with unknown relevance to humans).

MAK: 380 mg/m³, 200 ppm; peak limitation category: II(4); carcinogen category: 5; pregnancy risk group: C; germ cell mutagen group: 5**ENVIRONMENT**

Environmental effects of the substance have been adequately investigated, but no significant effects have been found.

NOTES

Ethanol consumption during pregnancy may adversely affect the unborn child.

Chronic ingestion of ethanol may cause liver cirrhosis and cancer.

ADDITIONAL INFORMATION**EC Classification**

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